

Heart Disease Prediction using Machine Learning

ARTIFICIAL INTELLIGENCE COURSE

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- **Introduction :**

Heart disease is one of the leading causes of death worldwide. Early prediction can help doctors take preventive actions and improve patient outcomes.

In this project, machine learning techniques are used to predict whether a patient has heart disease based on medical attributes such as age, cholesterol, blood pressure, and smoking status.

- **Abstract :**

This project applies machine learning techniques to build a predictive model for identifying the presence of heart disease in patients. The dataset contains multiple medical and lifestyle-related features, which are used as inputs to classification algorithms.

The dataset is first preprocessed to handle missing values, encode categorical variables, and ensure that the data is suitable for modeling. After preprocessing, multiple classification algorithms are applied, including Logistic Regression, Support Vector Machine (SVM), and Decision Tree.

The performance of each model is evaluated using different metrics such as accuracy, precision, recall, F1-score, and confusion matrix to determine the most effective model for this problem. Additionally, hyperparameter tuning is performed to analyze its impact on model performance and improve prediction accuracy.

The main objective of this project is to build a reliable predictive system that can assist in early detection of heart disease, which may help healthcare professionals make better and faster decisions.

- **Dataset Description :**

- Dataset contains:

- I. Train data: 224 row each row consist of 16 independent features.
- II. Test data: 56 row each row consist of 16 independent features.
- III. Target : (Heart Disease) has 2 classes (Yes (Has heart disease) , No (Hasn't heart disease))
- IV. Features : Age,Gender,Chest pain type,BP,Cholesterol,FBS over120,EKG results,Max HR,Exercise angina,ST depression,Slope of ST ,Number of vessels fluro,Thallium, work_type,smoking_status.

- **Data Preprocessing :**

- **Data Cleaning:**

The datasets were initially loaded, and the **id** column was removed as it does not contribute to the prediction process. Additionally, duplicate records were identified and removed to improve data quality. A total of **9 duplicates** were removed from the training dataset and **1 duplicate** from the testing dataset.

- **Target Encoding:**

The target variable "**Heart Disease**" was converted into a numerical format using **Label Encoding**, where:

yes --> 1

No --> 0

This transformation is necessary as ML models require numerical inputs.

- **Handling Inconsistent Categories:**

The **work_type** feature contained inconsistent or illogical values such as "*children*". These values were replaced with "*Never_worked*" to maintain consistency and ensure meaningful interpretation of the feature.

- **Handling Missing Values:**

Missing values were handled using statistical measures computed from the training dataset:

- **Age** -> filled using the *median*
- **Gender** -> filled using the *mode*
- **work_type** -> filled using the *mode*
- **smoking_status** -> filled with "*Unknown*"

The same values were applied to the testing dataset to avoid data leakage and maintain consistency.

- **Outlier Treatment (IQR Method):**

Outliers in numerical features (Age, BP, Cholesterol, Max HR, ST depression) were detected using the Interquartile Range (IQR) method. Values falling outside the lower and upper bounds were capped (clipped) to reduce the influence of extreme values and improve model stability.

➤ **Encoding Categorical Variables:**

Categorical features (Gender, work_type, smoking_status) were transformed using **One-Hot Encoding**. To ensure consistency between datasets, the training and testing sets were aligned to have identical feature structures after encoding.

➤ **Feature Scaling:**

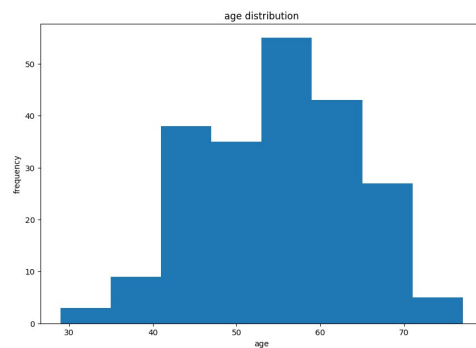
Numerical features were standardized using StandardScaler.

The scaler was fitted on the training data and applied to the testing data.

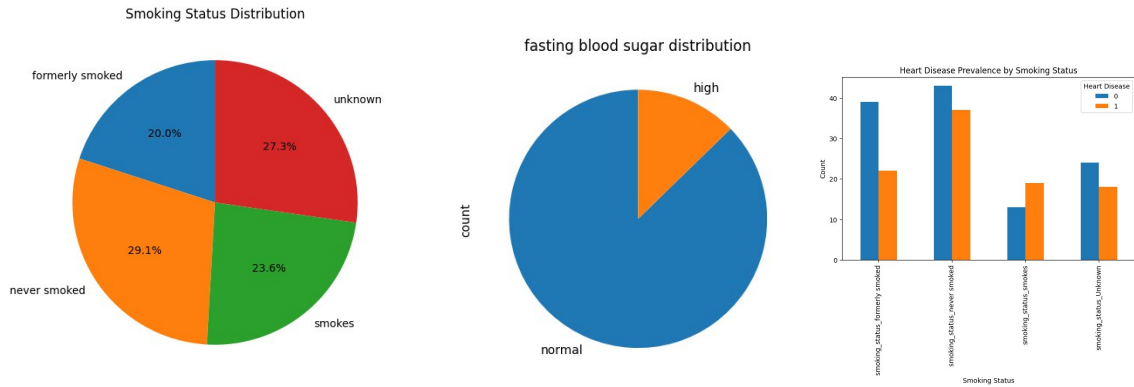
• **Visualization :**

➤ The dataset was explored to understand demographic and lifestyle patterns.

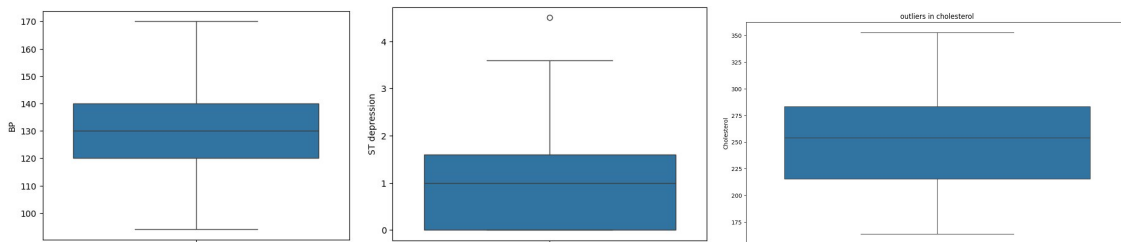
➤ A histogram of age revealed the most common age groups within the population



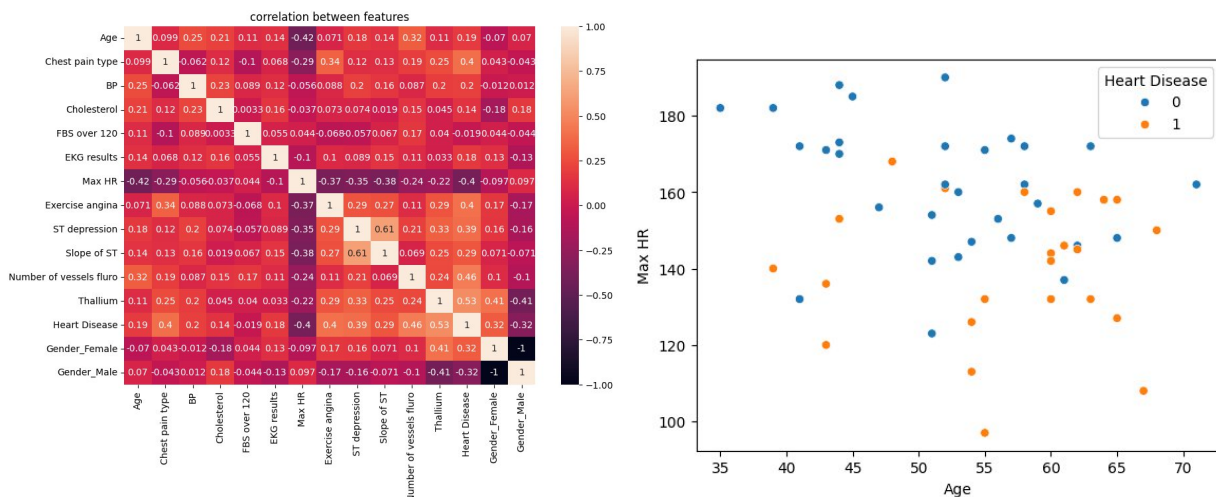
➤ Pie charts were used to illustrate the distribution of smoking status and fasting blood sugar (FBS), providing insight into key lifestyle risk factors. Additionally, a grouped bar chart was used to analyze heart disease prevalence across different work types, highlighting potential occupational risk pattern



- Box plots were used to detect outliers and examine the distribution of clinical variables such as cholesterol, blood pressure, and ST depression. A count plot of the number of vessels (fluro) confirmed its strong association with heart disease. Comparative box plots showed that patients with heart disease tend to have higher median cholesterol and ST depression values compared to healthy individuals.



- A correlation heatmap was used to measure relationships between numerical features, showing that age has a notable influence on several clinical indicators. Furthermore, a scatter plot between age and maximum heart rate demonstrated a clear negative correlation, indicating that maximum heart rate decreases with age.



- **SVM Heart Disease Classification Report :**

1. Model Description

A Support Vector Machine (SVM) classifier was developed to predict whether a patient has heart disease or not. The model was trained using a standard clinical dataset and evaluated on a separate test set. To optimize performance, different kernel types (RBF and Linear) and various regularization parameters (C values) were tested to analyze their effect on classification accuracy and generalization.

2. Comparison of Results

Kernel	C value	Accuracy	Precision	Recall	F1-score
RBF	0.1	0.80	0.76	0.79	0.78
RBF	1	0.836	0.826	0.792	0.809
RBF	10	0.782	0.773	0.708	0.739
Linear	0.1/1/10	0.818	0.792	0.792	0.792

3. Confusion Matrix Analysis

Model	Actual Condition	Predicted: Healthy	Predicted: Disease
RBF (C=0.1)	No Heart Disease	25 (TN)	6 (FP)
	Heart Disease	5 (FN)	19 (TP)
RBF (C=1)	No Heart Disease	27 (TN)	4 (FP)
	Heart Disease	5 (FN)	19 (TP)
RBF (C=10)	No Heart Disease	26 (TN)	5 (FP)
	Heart Disease	7 (FN)	17 (TP)
Linear Kernel	No Heart Disease	26 (TN)	5 (FP)
	Heart Disease	5 (FN)	19 (TP)

4. Technical Analysis & Hyperparameters

- Kernel Selection: The RBF kernel outperformed the Linear kernel, indicating that the decision boundary for heart disease features is non-linear in nature.

- Regularization (C):

- At $C=0.1$, the model was too simple (underfitting).
- At $C=10$, accuracy dropped to 78.2%, indicating that the model began to overfit the training noise, losing its ability to generalize to new data.

5. Conclusion

The SVM model achieved good performance, with the best accuracy (83.6%) obtained using RBF kernel with $C=1$.

• Decision Tree Heart Disease Classification Report :

1. Model Description

A Decision Tree classifier was developed to predict whether a patient has heart disease or not. The model was trained on a labeled clinical dataset and evaluated on a separate test set. Different values of the hyperparameter (`max_depth`) were tested to study their effect on performance and generalization.

2. Comparison of Results

Model	max_depth	Accuracy	Precision	Recall	F1-score
1	3	0.855	0.864	0.792	0.826
2	5	0.836	0.857	0.750	0.800
3	10	0.782	0.750	0.750	0.750
4	None	0.782	0.750	0.750	0.750

3. Confusion Matrix Analysis

Model	max_depth	TN	FP	FN	TP
1	3	28	3	5	19
2	5	28	3	6	18
3	10	26	5	5	19
4	None	26	5	4	20

4. Technical Analysis& Hyperparameters

The performance of the Decision Tree model depends on the max_depth parameter. A smaller depth may lead to underfitting, while a larger depth may lead to overfitting. Model 1 achieved the best balance between precision and recall, while deeper models showed reduced generalization.

5. Conclusion

The Decision Tree model achieved good performance in predicting heart disease. The best results were obtained with max_depth=3, which provided the highest accuracy and balanced performance.

- **Random Forest Heart Disease Classification Report :**

- 1. Model Description**

A Random Forest Classifier was developed to predict whether a patient has heart disease or not. The model used is the Random Forest Classifier, which combines multiple decision trees to improve prediction accuracy.

Each tree is trained on different parts of the data, and the final prediction is made using majority voting. This helps reduce overfitting and improves generalization.

- 2. Comparison of Results**

model_id	n_estimators	max_depth	Accuracy	Precision	Recall	F1_Score
1	50	5	0.85	0.86	0.79	0.82
2	50	10	0.85	0.86	0.79	0.82
3	50	None	0.85	0.86	0.79	0.82
4	100	5	0.85	0.86	0.79	0.82
5	100	10	0.87	0.86	0.83	0.85
6	100	None	0.87	0.86	0.83	0.85
7	200	5	0.89	0.9	0.83	0.86
8	200	10	0.89	0.9	0.83	0.86
9	200	None	0.89	0.9	0.83	0.86

- 3. Confusion Matrix Analysis**

model_id	n_estimators	max_depth	TN	FP	FN	TP
1	50	5	28	3	5	19
2	50	10	28	3	5	19
3	50	None	28	3	5	19
4	100	5	28	3	5	19
5	100	10	28	3	4	20
6	100	None	28	3	4	20
7	200	5	29	2	4	20
8	200	10	29	2	4	20
9	200	None	29	2	4	20

4. Technical Analysis& Hyperparameters

several Random Forest models were trained with different hyperparameters. The following observations were made:

- **n_estimators:** Increasing the number of trees (n) to 200 leads to more stable results and higher accuracy (0.891).
- **max_depth:** A depth of 5 or 10 was sufficient to achieve optimal performance without excessive complexity.
- **Model ID 7:** This model (n=200, depth=5) is selected as the best configuration, providing the highest accuracy and a balanced confusion matrix.

5. Conclusion

The Random Forest model was successful in classifying heart disease with high performance. After evaluating nine different models, we conclude the following:

- **Optimal Performance:** Model **ID 7** achieved the highest accuracy of **0.891**.
- **Best Configuration:** Using **200 trees** (n_estimators) with a **max_depth of 5** provided the most stable and balanced results.
- **Final Outcome:** The model effectively minimized errors, making it a reliable tool for this classification task.

Logistic Regression Heart Disease Classification Report :

1. Model Description

Heart Disease Prediction Using Logistic Regression

The Logistic Regression algorithm has been used to predict if the patient has heart disease or not. The algorithm uses the sigmoid function to estimate the probability of the target class.

Various hyperparameters such as a regularizer (C), penalty type (L1 and L2), have been used to test their effects on performance.

2. Comparison of Results

Penalty C	value	Accuracy	Precision	Recall	F1-score
L2	0.001	0.730	0.921	0.437	0.592
L2	0.1	0.828	0.853	0.760	0.798
L2	1	0.819	0.825	0.770	0.790
L2	10	0.805	0.802	0.760	0.776
L1	0.001	0.553	0.000	0.000	0.000
L1	0.1	0.763	0.736	0.758	0.738
L1	1	0.805	0.804	0.749	0.773
L1	10	0.809	0.809	0.760	0.780

3. Confusion Matrix Analysis

Actual Condition	Predicted: Healthy	Predicted: Disease
No Heart Disease	29 (TN)	2 (FP)
Heart Disease	5 (FN)	19 (TP)

4. Technical Analysis & Hyperparameters

- L2 regularization outperformed L1 in general performance, mainly in Accuracy and F1-score, suggesting that the former worked better with this dataset.

Effect of C:

- When $C = 0.001$, the model had too high bias and variance and therefore performed poorly, particularly when applying L1 regularization, resulting in Recall and F1-scores equal to 0.
- When $C = 0.1$, the classifier reached its optimal state with the lowest level of bias and variance. In this case, the best results in Accuracy (82.8%) and F1-score (79.8%) have been obtained.
- The increase in the C value up to 1 and 10 resulted in a slight decrease in performance; it might be due to the overfitting issue caused by an increased variance.

5. Conclusion

The Logistic Regression model demonstrated strong and stable performance in predicting heart disease. the model with $C=0.1$ and L2 regularization achieved the best performance, with

- Average Accuracy = 82.8%
- Average F1-score = 79.8%

Naive Bayes Heart Disease Classification Report

1-model description:

In this project, the Gaussian Naive Bayes model was chosen as the classification algorithm. This type best suits our data for the following reasons:

- It is designed for continuous numerical features, which matches the medical dataset used.
- It is fast, simple, and effective for binary classification tasks such as heart disease prediction.
- It performs well even with relatively small datasets and independent feature assumptions.

2- Feature Selection using SelectKBest:

To improve model performance, feature selection was applied using SelectKBest, this method :
Evaluates each feature individually, Selects the top k most relevant features that contribute most to prediction accuracy

Different values of k were tested using cross-validation, and the best-performing number of features was selected to optimize model performance while reducing noise and redundancy.(k=15)

3- Hyperparameter(var_smoothing):

The **var_smoothing** parameter is a hyperparameter used in Gaussian Naive Bayes to improve numerical stability.it works by adding a small value to the variance of features, preventing issues such as: Very small variance values ,Division by zero errors ,Numerical instability in probability calculations.

This parameter was tested with different values to ensure optimal model stability and performance.

4- Model Evaluation Comparison :

Evaluation Metric	Training	Test Set
Accuracy	0.84	0.81
Precision	0.84	0.77
Recall	0.81	0.83
F1-Score	0.82	0.80

Cross-validation results represent model performance on training data using multiple folds, while the test set results reflect the model's ability to generalize to unseen data.

The close performance between training and testing results indicates that the model generalizes well and does not suffer from overfitting.

5-Confusion Matrix :

Actual Condition	Predicted:	
	Healthy	Disease
No Heart Disease	25 (TN)	6 (FP)
Heart Disease	4 (FN)	20 (TP)

The confusion matrix indicates that the model performs well in distinguishing between positive and negative cases, with a relatively low number of misclassifications. The number of false negatives is small, which is particularly important in medical diagnosis, as missing a diseased patient can have serious consequences. Overall, the model shows a good balance

6-conclusion :

in this project, we used a Gaussian Naive Bayes model to predict heart disease.

After selecting the most important features (best $k = 15$) and tuning the model, it gave good and stable results on both training and test data. The model was able to correctly classify most cases and showed balanced performance with an F1-score around 0.80.

Overall, the model works well and can be used as a good starting point for heart disease prediction.

Final Model Comparison and Selection:

Model	Accuracy	F1-score
SVM	0.836	0.809
Decision Tree	0.855	0.826
Logistic Regression	0.828	0.798
Naïve Bayes	0.81	0.80
Random Forest	0.89	0.86

After evaluating multiple machine learning models, including Support Vector Machine, Decision Tree, Logistic Regression, Naïve Bayes, and Random Forest, a comparison was conducted based on performance metrics such as **accuracy** and **F1-score**.

Among all tested models, **the Random Forest model** achieved the best performance, with the **highest accuracy (0.89) and F1-score (0.86)**. It also showed a well-balanced confusion matrix with fewer misclassifications compared to other models.

Therefore, **Random Forest was selected as the final and most effective model for heart disease prediction in this project.**